(4.1)

ⓐ $N=20$, $\sum Y_i^2=7825.94$, $\bar{Y}=19.21$, $SSR=375.47$

$$\Rightarrow R^2=\frac{SSR}{SSTO}=\frac{375.47}{445.458}\approx 0.8429 \#$$

$$\sum(Y_i-\bar{Y})^2=\sum Y_i^2-N\bar{Y}^2=7825.94-20\times 19.21^2=445.458$$

ⓑ

$R^2=0.7911$, $SST=725.94$, $N=20$

$$\Rightarrow \hat{\sigma}^2=\frac{\sum \hat{e}_i^2}{N-2}=\frac{\sum(Y_i-\hat{Y}_i)^2}{N-2}=\frac{SSE}{N-2}=\frac{151.6489}{20-2}=8.4249 \#$$

其中 $R^2=0.7911=1-\frac{SSE}{SST}=1-\frac{SSE}{725.94} \rightarrow SSE \doteq 151.6489$

ⓒ $\sum(Y_i-\bar{Y})^2=631.63$, $\sum \hat{e}_i^2=182.85$

$$\Rightarrow R^2=\frac{SSR}{SSTO}=\frac{631.63-182.85}{631.63}=0.7105 \#$$

(4.3)

ⓐ $\hat{y}=1.2+0.8X$

$\rightarrow \hat{y}_0=1.2+0.8\times 4=4.4 \#$

ⓑ $f=\beta_1+\beta_2X_0+e_0-(b_1+b_2X_0)$

$$\widehat{Var}(f)=\hat{\sigma}^2\left[1+\frac{1}{N}+\frac{(X_0-\bar{X})^2}{\sum_{i=1}^{N}(X_i-\bar{X})^2}\right]=1.2\left[1+\frac{1}{5}+\frac{(4-1)^2}{10}\right]=2.52$$

$$\hat{\sigma}^2=\frac{\sum \hat{e}_i^2}{n-2}=\frac{3.6}{5-2}=1.2$$

$$\bar{X}=\frac{3+2+1+(-1)+0}{5}=1$$

$$se(f)=\sqrt{2.52}\approx 1.5875 \#$$

ⓒ a 95% prediction interval for y given $X_0=4$

$$\Rightarrow \hat{y}_0 \pm t_c^* se(f)=4.4\pm 3.182\times 1.5875 \Rightarrow [-0.6514, 9.4514] \#$$

$t_{0.025}(3)=3.182$

ⓓ a 99% prediction interval for y given $X_0=4$

$$\Rightarrow \hat{y}_0 \pm t_{0.005}(3) * se(f)=4.4\pm 5.841 * 1.5875 \Rightarrow [-4.8726, 13.6726] \#$$

(e)

$\hat{y}_0 = 1.2 + 0.8 \times 1 = 2$

$f = \beta_0 + \beta_1 x_0 + e_0 - (b_0 + b_1 x_0)$

$\widehat{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{5} (x_i - \bar{x})^2} \right] = 1.2 \left[1 + \frac{1}{5} + \frac{0^2}{10} \right] = 1.44$

$\hat{\sigma}^2 = \frac{\sum_{i=1}^{5} \hat{e}_i^2}{n-2} = 1.2$

$se(f) = \sqrt{\widehat{var}(f)} = 1.2$

a 95% prediction interval for y given $x = \bar{x}$

$\rightarrow \hat{y}_0 \pm t_{0.025}(3) * se(f) = 2 \pm 3.182 \times 1.2 \Rightarrow [-1.8184, 5.8184]$ #

預測點 x_0 離樣本平均值 $\bar{x}$ 越遠，預測的不確定性越大，因此區間越寬；在 $x = \bar{x}$ 時，預測區間最窄。

(c) the width of the interval $\rightarrow 10.1028 > 7.6368 \leftarrow$ (e) the width of the interval

(4.25)

#a

The estimated log-linear regression model is expressed as ln(Price) = 4.3939 + 0.0360(SQFT). The intercept of 4.3939 represents the predicted log-price when the house size is zero; however, this value holds little practical significance as it represents an extrapolation outside the observed data range where a zero-square-foot house does not exist. The slope coefficient for SQFT is 0.0360, indicating that for every additional 100 square feet (1 unit), the house price is expected to increase by approximately 3.60%. This relationship is even statistically significant at the 1% level ($p < 0.01$). Furthermore, the elasticity at the mean is calculated to be 0.983, suggesting a near-unitary relationship where house prices move almost proportionately with the physical size of the property."

```
[a] Log-Linear Model Summary:
> print(summary(model_a))

Call:
lm(formula = log(price) ~ sqft, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.32528 -0.19199  0.00122  0.21678  0.86609 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408 
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16

> # 彈性 (Elasticity at means): beta2 * mean(sqft)
> elasticity_a <- coef(model_a)["sqft"] * mean(df$sqft)
> cat("Elasticity at mean (Log-Linear):", elasticity_a, "\n")
Elasticity at mean (Log-Linear): 0.9833701
```

#b

The estimated log-log regression model is expressed as ln(Price) = 2.0497 + 1.0248 ln(SQFT). The intercept of 2.0497 represents the predicted log-price when ln(SQFT) = 0 (equivalent to a house size of 1 unit or 100 square feet). This back-transforms to a base price of approximately $7,765 {(e^2.0497) * 1000}; however, this value holds little practical significance as it represents an extrapolation far below the typical house sizes found in the dataset. The slope coefficient for ln(SQFT) is 1.0248, which directly represents the constant elasticity of price with respect to size. This implies that for every 1% increase in square footage, the house price is expected to increase by approximately 1.02%. This relationship is statistically significant at the 1% level ($p < 0.01$). This elasticity slightly exceeding unity suggests that in this model, there are subtle 'increasing returns' to size, where larger houses command a slightly higher proportional price premium."

```
[b] Log-Log Model Summary:
> print(summary(model_b))

Call:
lm(formula = log(price) ~ log(sqft), data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16

> # 彈性 (Elasticity): 在 Log-Log 中即為斜率係數
> elasticity_b <- coef(model_b)["log(sqft)"]
> cat("Elasticity (Log-Log):", elasticity_b, "\n")
Elasticity (Log-Log): 1.024828 
> 
```

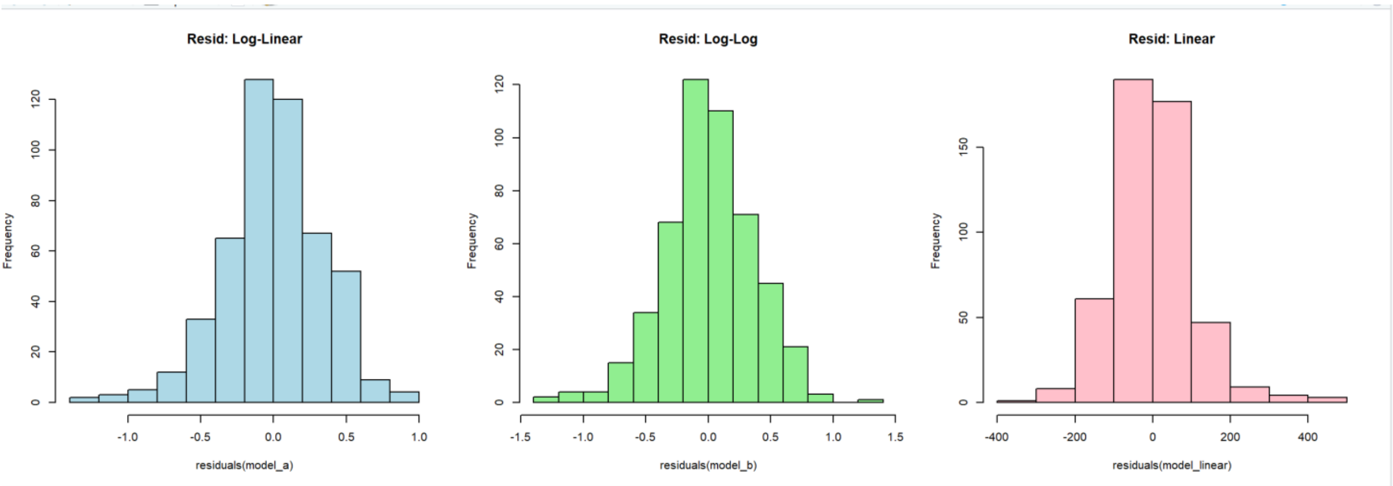
#c

Based on the results, the Log-Linear model (a) achieves the highest explanatory power with a Generalized R^2 of 0.6622, followed by the Log-Log model (0.6445) and the Linear model (0.6413). This indicates that the Log-Linear specification provides the best fit for the data, explaining approximately 66.22% of the variation in house prices. The superior performance of the log-linear model suggests that house prices in this dataset are better modeled by a constant percentage growth rate relative to size rather than a strictly linear or constant elasticity relationship."

```
[c] R-squared Comparison:
> cat("Linear R2:", summary(model_linear)$r.squared, "\n")
Linear R2: 0.6413167 
> cat("Log-Linear Generalized R2:", gen_r2_a, "\n")
Log-Linear Generalized R2: 0.6621612 
> cat("Log-Log Generalized R2:", gen_r2_b, "\n")
Log-Log Generalized R2: 0.6445084 
```

#d

The Jarque-Bera (JB) test was conducted to evaluate the normality of residuals for all three models. The null hypothesis states that the residuals are normally distributed. According to the results, the p-values for the Log-Linear (1.61*10^{-6}), Log-Log (0.00079), and Linear (0) models are all significantly lower than the 0.05 threshold. Therefore, we reject the null hypothesis for all three specifications, indicating that none of the models have strictly normally distributed residuals.

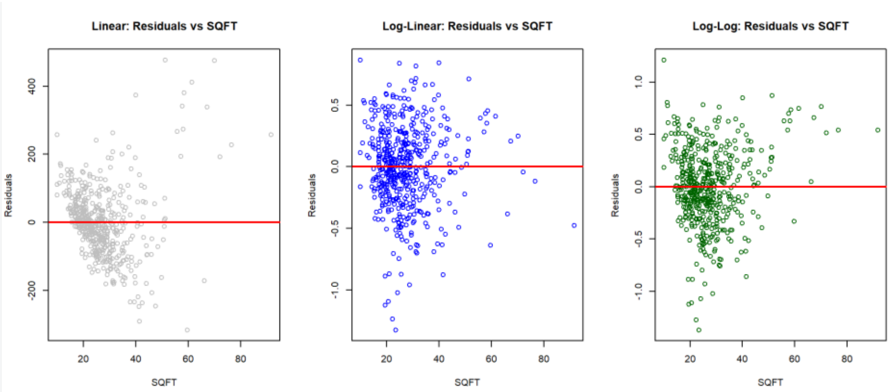



```
>
> # 模型誤差是否符合常態分佈(Jarque Bera Test)
> # 虛無假設是符合常態，故P-Value>alpha時不可拒絕虛無
> jb_a <- jarque.bera.test(residuals(model_a))
> jb_b <- jarque.bera.test(residuals(model_b))
> jb_linear <- jarque.bera.test(residuals(model_linear))
> cat("\n[d] Jarque-Bera Test p-values:\n")

[d] Jarque-Bera Test p-values:
> cat("Log-Linear:", jb_a$p.value, "| Log-Log:", jb_b$p.value, "| Linear:", jb_linear$p.value, "\n")
Log-Linear: 1.609651e-06 | Log-Log: 0.0007932632 | Linear: 0
```

#e

The residual plots for the Linear, Log-Linear, and Log-Log models illustrate the relationship between the residuals and the independent variable (SQFT). In the Linear model (left), the residuals exhibit a distinct 'fan-shaped' pattern, where the spread of residuals increases significantly as SQFT grows. This is a clear indication of heteroskedasticity, suggesting that the model's error variance is not constant. In contrast, the Log-Linear (center) and Log-Log (right) models show a much more uniform and random distribution of residuals around the zero line. By transforming the dependent variable into logs, these models effectively stabilize the variance, mitigating the heteroskedasticity issue found in the linear specification. However, a few outliers remain visible in all three plots, which aligns with the non-normality results from the Jarque-Bera test



#f #g

```
[Model C - Linear] 預測值與區間:
> print(pred_linear)
       fit      lwr      upr
1 246.4557 44.27727 448.6341
```

```
[Model A - Log-Linear] 預測值與區間 (已還原):
> print(pred_a_price)
       fit      lwr      upr
1 214.2336 109.7685 418.1167
```

```
[Model B - Log-Log] 預測值與區間 (已還原):
> print(pred_b_price)
       fit      lwr      upr
1 227.5386 111.1406 465.8406
>
```

#h

Based on a comprehensive evaluation of the three models, I recommend the Log-Linear model (a) as the preferred specification for estimating house prices in Collegetown. My recommendation is supported by the following two criteria:

1. Superior Explanatory Power: The Log-Linear model achieves the highest Generalized R^2 of 0.6622, indicating it explains a larger portion of the variation in house prices compared to the Log-Log (0.6445) and Linear (0.6413) models.
2. Structural Stability: Unlike the Linear model, which suffers from significant heteroskedasticity (as evidenced by the fan-shaped residual plot in part e), the Log-Linear model successfully stabilizes the error variance through its semi-logarithmic transformation.